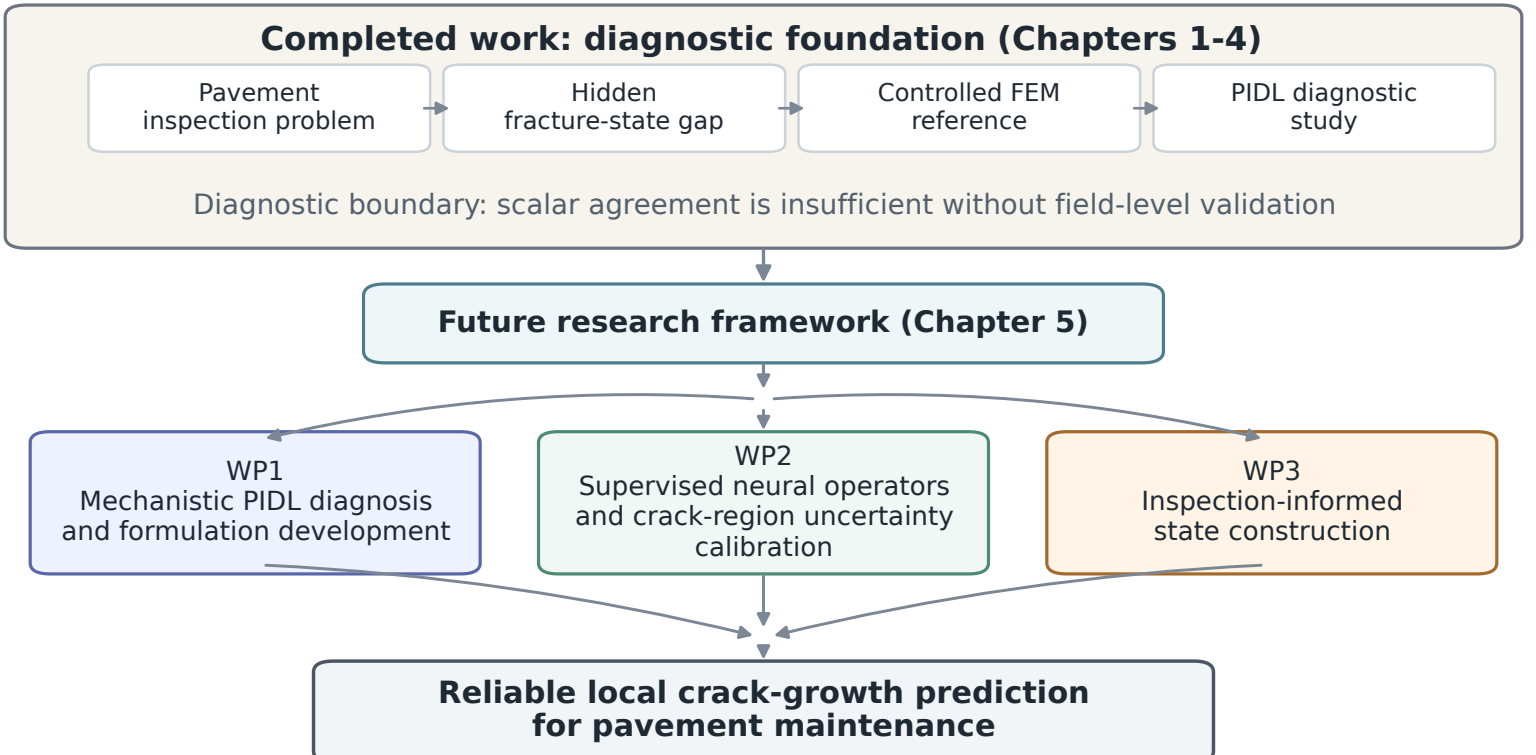
Completed work: diagnostic foundation (Chapters 1-4)
Pavement inspection problem
Hidden fracture-state gap
Controlled FEM reference
PIDL diagnostic study
Diagnostic boundary: scalar agreement is insufficient without field-level validation
Future research framework (Chapter 5)
WP1
Mechanistic PIDL diagnosis and formulation development
WP2
Supervised neural operators and crack-region uncertainty calibration
WP3
Inspection-informed state construction
Reliable local crack-growth prediction for pavement maintenance